LAPORAN PRAKTIKUM

JARINGAN KOMPUTER 2

Ujian Tengah Semester



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PROGRAM STUDI INFORMATIKA

UNIVERSITAS KATOLIK MUSI CHARITAS

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MATERI : Ujian Tengah Semester

PERTEMUAN KE : 15 dan 16

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1. Anda sebagai administrator jaringan memiliki jaringan dengan alamat network 200.20.2.0/24. Jaringan ini perlu dibagi menjadi beberapa subnet untuk mengakomodasi departemen-departemen berikut:
   - Departemen Sales: Membutuhkan 40 host.
   - Departemen IT: Membutuhkan 110 host.
   - Departemen HR: Membutuhkan 40 host.
   - Departemen Finance: Membutuhkan 20 host.

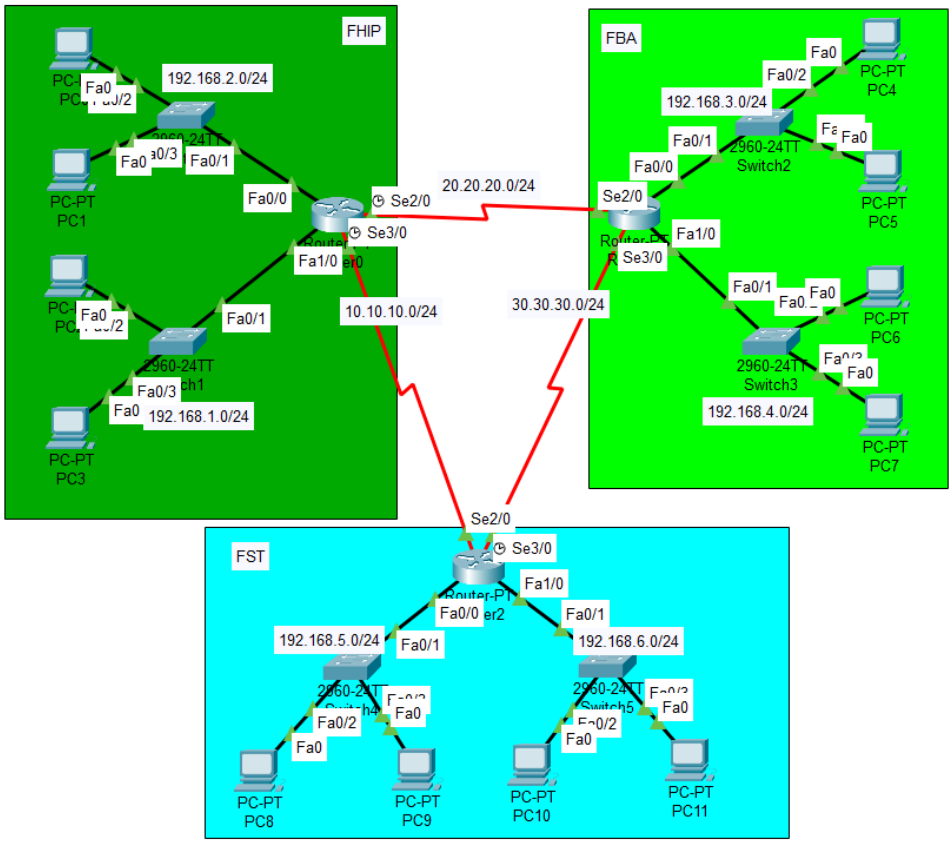
   Tentukan subnet mask yang sesuai untuk masing-masing departemen dan tentukan alamat network serta broadcast address untuk setiap subnet (hitung pakai Excel).

subnet	2^0=1
host persubnet	2^8-2=254
blok subnet	256-0=256

subnet	200.20.2.0
host pertama	200.20.2.1
host terakhir	200.20.2.254
broadcast	200.20.2.255

ALAMAT IP	IT	SALES	HR	FINANCE
Network/sub	200.20.2.0/25	200.20.2.128/26	200.20.2.192/26	200.20.2.224/27
host pertama	200.20.2.1/25	200.20.2.129/26	200.20.2.193/26	200.20.2.225/27
host terakhir	200.20.2.126/25	200.20.2.190/26	200.20.2.254/26	200.20.2.254/27
broadcast	200.20.2.127/25	200.20.2.191/26	200.20.2.255/26	200.20.2.255/27

2. Konfigurasi topologi jaringan tersebut menggunakan routing OSPF single area



- Konfigurasi IP address di Router 0 FHIP

  Router>en
  Router#conf t
  Router(config)#int fa0/0
  Router(config-if)#no sh
  Router(config-if)#ip add 192.168.2.1 255.255.255.0
  Router(config)#int fa1/0
  Router(config-if)#no sh
  Router(config-if)#ip add 192.168.1.1 255.255.255.0
  Router(config)#int se2/0
  Router(config-if)#no sh
  Router(config-if)#ip add 20.20.20.1 255.255.255.0
  Router(config)#int se3/0
  Router(config-if)#no sh
  Router(config-if)#ip add 10.10.10.1 255.255.255.0

- Konfigurasi OSPF Router 0

  Router(config)#router ospf 1
  Router(config-router)#net 192.168.1.0 0.0.0.255 area 0
  Router(config-router)#net 192.168.2.0 0.0.0.255 area 0

Router(config-router)#net 10.10.10.0 0.0.0.255 area 0
Router(config-router)#net 20.20.20.0 0.0.0.255 area 0

- Konfigurasi IP address di Router 1 FBA

Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config)#int fa1/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.4.1 255.255.255.0
Router(config)#int se2/0
Router(config-if)#no sh
Router(config-if)#ip add 20.20.20.2 255.255.255.0
Router(config)#int se3/0
Router(config-if)#no sh
Router(config-if)#ip add 30.30.30.1 255.255.255.0

- Konfigurasi OSPF Router 1

Router(config)#router ospf 1
Router(config-router)#net 192.168.3.0 0.0.0.255 area 0
Router(config-router)#net 192.168.4.0 0.0.0.255 area 0
Router(config-router)#net 30.30.30.0 0.0.0.255 area 0
Router(config-router)#net 20.20.20.0 0.0.0.255 area 0

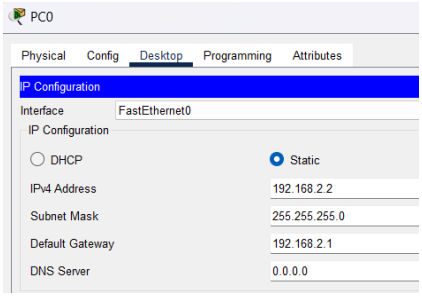
- Konfigurasi IP address di Router 2 FST

Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.5.1 255.255.255.0
Router(config)#int fa1/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.6.1 255.255.255.0
Router(config)#int se2/0
Router(config-if)#no sh
Router(config-if)#ip add 10.10.10.2 255.255.255.0
Router(config)#int se3/0
Router(config-if)#no sh
Router(config-if)#ip add 30.30.30.2 255.255.255.0

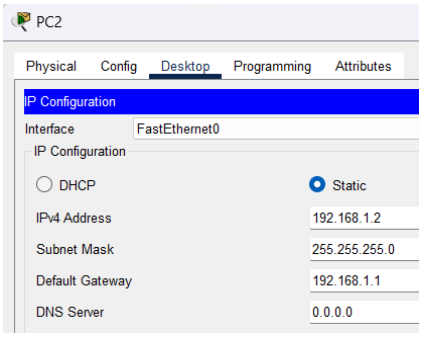
- Konfigurasi OSPF Router 2

  Router(config)#router ospf 1
  Router(config-router)#net 192.168.5.0 0.0.0.255 area 0
  Router(config-router)#net 192.168.6.0 0.0.0.255 area 0
  Router(config-router)#net 10.10.10.0 0.0.0.255 area 0
  Router(config-router)#net 30.30.30.0 0.0.0.255 area 0

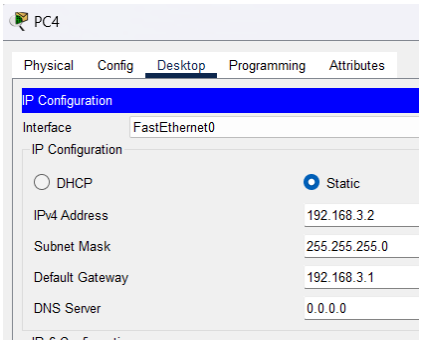
- IP PC 0



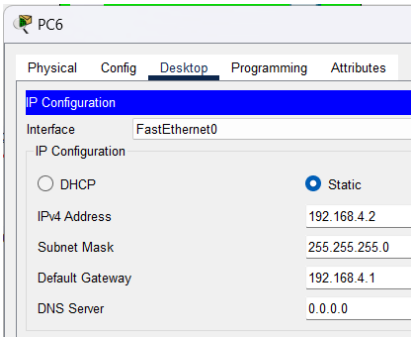
- IP PC 2



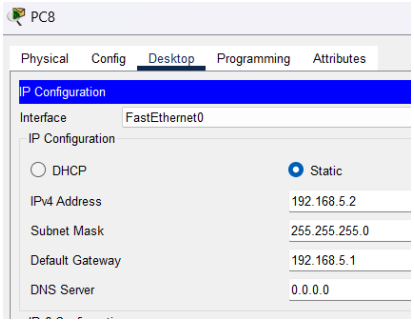
- IP PC 4



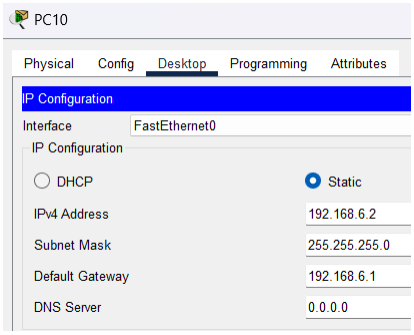
- IP PC 6



- IP PC 8



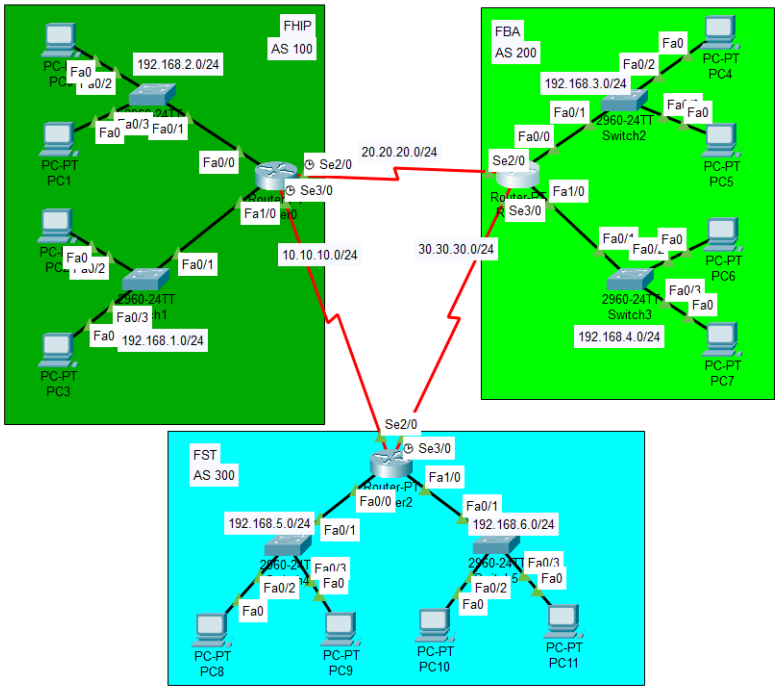
- IP PC 10



- Percobaan Koneksi

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC10	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC6	PC10	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)	(delete)

3. Konfigurasi topologi jaringan tersebut menggunakan satu AS



- Konfigurasi IP address di Router 0 FHIP

  Router>en
  Router#conf t
  Router(config)#int fa0/0
  Router(config-if)#no sh
  Router(config-if)#ip add 192.168.2.1 255.255.255.0
  Router(config)#int fa1/0
  Router(config-if)#no sh
  Router(config-if)#ip add 192.168.1.1 255.255.255.0
  Router(config)#int se2/0
  Router(config-if)#no sh
  Router(config-if)#ip add 20.20.20.1 255.255.255.0
  Router(config)#int se3/0
  Router(config-if)#no sh
  Router(config-if)#ip add 10.10.10.1 255.255.255.0

- Konfigurasi BGP Router 0

  Router(config)#router bgp 100
  Router(config-router)#net 192.168.1.0 mask 255.255.255.0
  Router(config-router)#net 192.168.2.0 mask 255.255.255.0
  Router(config-router)#neighbor 20.20.20.2 remote-as 200
  Router(config-router)#neighbor 10.10.10.2 remote-as 300

- Konfigurasi IP address di Router 1 FBA

```
Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config)#int fa1/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.4.1 255.255.255.0
Router(config)#int se2/0
Router(config-if)#no sh
Router(config-if)#ip add 20.20.20.2 255.255.255.0
Router(config)#int se3/0
Router(config-if)#no sh
Router(config-if)#ip add 30.30.30.1 255.255.255.0
```

- Konfigurasi BGP Router 1

```
Router(config)#router bgp 200
Router(config-router)#net 192.168.3.0 mask 255.255.255.0
Router(config-router)#net 192.168.4.0 mask 255.255.255.0
Router(config-router)#neighbor 20.20.20.1 remote-as 100
Router(config-router)#neighbor 30.30.30.2 remote-as 300
```

- Konfigurasi IP address di Router 2 FST

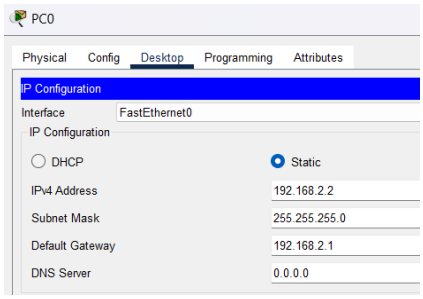
```
Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.5.1 255.255.255.0
Router(config)#int fa1/0
Router(config-if)#no sh
Router(config-if)#ip add 192.168.6.1 255.255.255.0
Router(config)#int se2/0
Router(config-if)#no sh
Router(config-if)#ip add 10.10.10.2 255.255.255.0
Router(config)#int se3/0
Router(config-if)#no sh
Router(config-if)#ip add 30.30.30.2 255.255.255.0
```

- Konfigurasi BGP Router 2

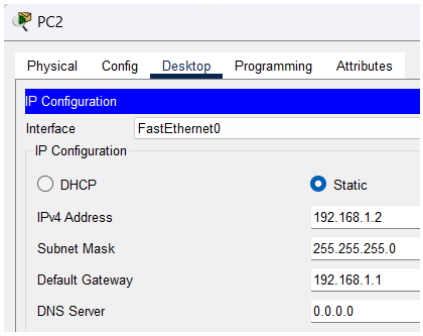
```
Router(config)#router bgp 300
Router(config-router)#net 192.168.5.0 mask 255.255.255.0
```

Router(config-router)#net 192.168.6.0 mask 255.255.255.0
Router(config-router)#neighbor 10.10.10.1 remote-as 100
Router(config-router)#neighbor 30.30.30.1 remote-as 200

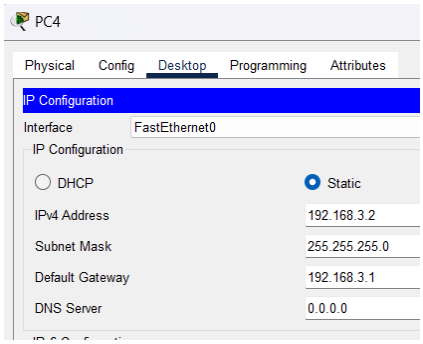
- IP PC 0



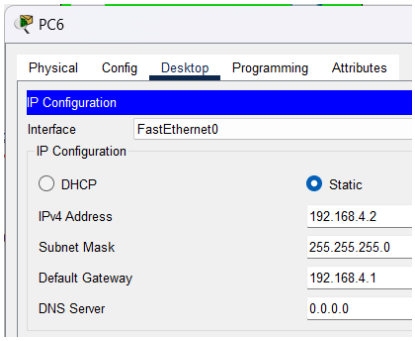
- IP PC 2



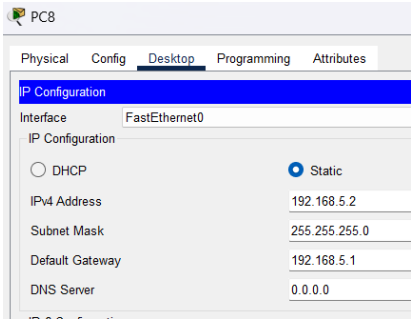
- IP PC 4



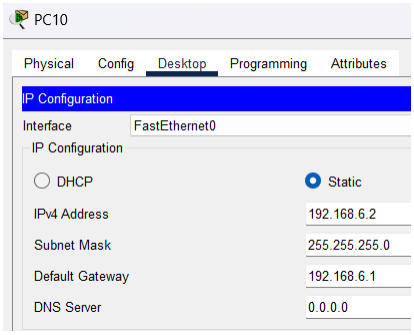
- IP PC 6



- IP PC 8



- IP PC 10



- Percobaan Koneksi

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC6	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC10	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC6	PC10	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	3	(edit)	(delete)